SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	M Dhanush Kumar	**Reg. No.:**	192525318
Section / Batch:		**Date:**	24/07/2026

Code of Conduct

I ____________(M Dhanush Kumar/192525318)__________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M Dhanush Kumar

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

My Assessments

Course: **CSA0305RA**

PSEUDOCODE WRITING TASK OPEN

CO5_AT1_Pseudocode Writing Task

0/5 answered

Attempt →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) OPEN ✓ Completed

DEBUGGING & OPTIMIZATION TASK

10/10 answered

Review →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) OPEN ✓ Completed

DEBUGGING & OPTIMIZATION TASK

10/10 answered

Review →

ENGINEER TO EXCEL

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack · Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q1 **Binary Search** 5 marks

Identify the error in the following snippet.
int binarySearch(int arr[], int n, int key) {
int low = 0, high = n - 1;
while(low < high) {
int mid = (low + high) / 2;
if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

int binarySearch(int arr[], int n, int key)
{
    int low = 0, high = n - 1;

    while (low <= high)
    {
        int mid = (low + high) / 2;

        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }

    return -1;
}

int main()
{
    int arr[] = {10, 20, 30, 40, 50};
    int n = 5;
    int key = 30;

    int result = binarySearch(arr, n, key);

    if (result != -1)
        printf("Element found at index %d", result);
    else
        printf("Element not found");

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Element found at index 2
```

Pending manual review

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

Q1	Binary Search	5 marks
Q2	Stack	5 marks
Q3	Stack	5 marks
Q4	Stack	5 marks
Q5	Stack · Balancing Symbols	5 marks
Q6	Singly Linked List	5 marks
Q7	Stack	5 marks
Q8	Linked List	5 marks
Q9	Linked List	5 marks
Q10	Queue	5 marks

10/10 answered

Q2 **Stack** 5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

int pop()
{
    if (top == -1)
    {
        printf("Underflow");
        return -1;
    }

    return stack[top--];
}

int main()
{
    stack[0] = 10;
    stack[1] = 20;
    top = 1;

    printf("Popped element: %d", pop());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Popped element: 20
```

Pending manual review

Submitted

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q3 Stack — 5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top;

void init()
{
    top = -1;
}

int main()
{
    init();

    printf("Stack initialized");
    printf("\nTop = %d", top);

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Stack initialized
Top = -1
```

Pending manual review

Submitted

ENGINEER TO EXCEL

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- **Q1** Binary Search 5 marks
- **Q2** Stack 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Stack · Balancing Symbols 5 marks
- **Q6** Singly Linked List 5 marks
- **Q7** Stack 5 marks
- **Q8** Linked List 5 marks
- **Q9** Linked List 5 marks
- **Q10** Queue 5 marks

10/10 answered

Q4 Stack 5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
while(stack[top] != '(') {
postfix[j++] = pop();
}
pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <ctype.h>

char stack[100];
int top = -1;

void push(char ch)
{
    stack[++top] = ch;
}

char pop()
{
    return stack[top--];
}

int main()
{
    char expression[] = "(A+B)";
    char postfix[100];
    int i = 0, j = 0;
    char ch;

    while (expression[i] != '\0')
    {
        ch = expression[i];

        if (ch == '(')
        {
            push(ch);
        }
        else if (isalnum(ch))
        {
            postfix[j++] = ch;
        }
        else if (ch == ')')
        {
            while (top != -1 && stack[top] != '(')
            {
                postfix[j++] = pop();
            }

```

Test Input

stdin input (optional)

Output

```
Postfix: AB+
```

Pending manual review

ENGINEER TO EXCEL

ENGINEERING

MADINENI DHANUSH KUMAR
192525318RA

DS PROGRAMMING

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q5 Stack - Balancing Symbols 5 marks

In balancing Bracket, you identify the error.

```
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
pop();
} else {
return 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

char stack[100];
int top = -1;

void push(char ch)
{
    stack[++top] = ch;
}

char pop()
{
    return stack[top--];
}

int main()
{
    char expression[] = "{[()]}";
    int i;
    char ch;
    int balanced = 1;

    for (i = 0; expression[i] != '\0'; i++)
    {
        ch = expression[i];

        if (ch == '(' || ch == '[' || ch == '{')
        {
            push(ch);
        }
        else if (ch == ')' || ch == ']' || ch == '}')
        {
            if (top == -1)
            {
                balanced = 0;
                break;
            }

            if ((ch == ')' && stack[top] == '(') ||
                (ch == ']' && stack[top] == '[') ||
                (ch == '}' && stack[top] == '{'))
            {
```

Test Input

stdin input (optional)

Output

```
Balanced Brackets
```

Pending manual review

ENGINEER TO EXCEL

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack · Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q6 **Singly Linked List** 5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

struct Node
{
    int data;
    struct Node *next;
};

int main()
{
    struct Node *head = NULL;
    struct Node *temp;

    head = (struct Node *)malloc(sizeof(struct Node));
    head->data = 10;
    head->next = (struct Node *)malloc(sizeof(struct Node));

    head->next->data = 20;
    head->next->next = (struct Node *)malloc(sizeof(struct Node));

    head->next->next->data = 30;
    head->next->next->next = NULL;

    temp = head;

    while (temp != NULL)
    {
        printf("%d ", temp->data);
        temp = temp->next;
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
10 20 30
```

Pending manual review

Submitted

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q7 **Stack** 5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

int pop()
{
    if (top == -1)
    {
        printf("Underflow");
        return -1;
    }
    else
    {
        return stack[top--];
    }
}

int main()
{
    stack[++top] = 10;
    stack[++top] = 20;

    printf("Popped element: %d", pop());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Popped element: 20
```

Pending manual review

Submitted

ENGINEER TO EXCEL

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q8 **Linked List** 5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

struct Node
{
    int data;
    struct Node *next;
};

struct Node* reverseList(struct Node *head)
{
    struct Node *prev = NULL;
    struct Node *current = head;
    struct Node *next;

    while (current != NULL)
    {
        next = current->next;
        current->next = prev;
        prev = current;
        current = next;
    }

    return prev;
}

int main()
{
    struct Node *head;
    struct Node *second;
    struct Node *third;
    struct Node *temp;

    head = malloc(sizeof(struct Node));
    second = malloc(sizeof(struct Node));
    third = malloc(sizeof(struct Node));

    head->data = 10;
    head->next = second;

    second->data = 20;
    second->next = third;

    third->data = 30;
```

Test Input

stdin input (optional)

Output

```
30 20 10
```

Pending manual review

ENGINEER TO EXCEL

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack · Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q9 **Linked List** 5 marks

What is the error in the below code?
new.next = head
head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

struct Node
{
    int data;
    struct Node *next;
};

struct Node* reverseList(struct Node *head)
{
    struct Node *prev = NULL;
    struct Node *current = head;
    struct Node *next;

    while (current != NULL)
    {
        next = current->next;
        current->next = prev;
        prev = current;
        current = next;
    }

    return prev;
}

int main()
{
    struct Node *head;
    struct Node *second;
    struct Node *third;
    struct Node *temp;

    head = malloc(sizeof(struct Node));
    second = malloc(sizeof(struct Node));
    third = malloc(sizeof(struct Node));

    head->data = 10;
    head->next = second;

    second->data = 20;
    second->next = third;

    third->data = 30;
    third->next = NULL;
```

Test Input

stdin input (optional)

Output

```
30 20 10
```

Pending manual review

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack · Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q10 **Queue** 5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int Q[MAX];
int front = -1;
int rear = -1;

int DEQUEUE()
{
    if (front == -1)
    {
        printf("Underflow");
        return -1;
    }

    int data = Q[front];

    if (front == rear)
    {
        front = -1;
        rear = -1;
    }
    else
    {
        front++;
    }

    return data;
}

int main()
{
    Q[0] = 10;
    Q[1] = 20;

    front = 0;
    rear = 1;

    printf("Dequeued element: %d", DEQUEUE());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Dequeued element: 10
```

Pending manual review

ENGINEER TO EXCEL

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Postfix Expression
5 marks

10/10 answered

Q1 **Stack** 5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
int PEEK(int stack[], int top)
{
    return stack[top];
}
int main()
{
    int stack[] = {10,20,30};
    int top = 2;
    printf("%d", PEEK(stack, top));
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
30
```

Pending manual review

Submitted

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define MAX 5
int main()
{
    int queue[MAX];
    int front = 0;
    int rear = 0;
    int vehicle = 10;
    if ((rear + 1)% MAX == front)
{
    printf("Queue full");
}
else
{
    rear = (rear + 1)% MAX;
    queue[rear] = vehicle;
    printf("vehicle added");
}
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
vehicle added
```

Pending manual review

Submitted

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

Q3 **Stack** 5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define MAX 5
int main()
{
    int top = MAX - 1;
    if (top == MAX - 1)
    {
        printf("overflow");
       }
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
overflow
```

Pending manual review

Submitted

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q4 **Stack** 5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
int main()
{
    int top = 0;
    if (top == 0)
    {
        printf("underflow");
    }
    else
    {
        top--;
        printf("element popped");
    }
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
underflow
```

Pending manual review

Submitted

Search

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

Q5 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.
void deleteNode(Node* del) {
if (del->prev != NULL)
del->prev->next = del->next;
if (del->next != NULL)
del->next->prev = del->prev;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

typedef struct Node
{
    int data;
    struct Node *prev;
    struct Node *next;
} Node;

void deleteNode(Node *del)
{
    if (del->prev != NULL)
    {
        del->prev->next = del->next;
    }

    if (del->next != NULL)
    {
        del->next->prev = del->prev;
    }
}

int main()
```

Test Input

stdin input (optional)

Output

```
Node deleted successfully
```

Pending manual review

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q6 **Queue** 5 ma

Identify the error in the Queue enqueue logic.
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#define MAX 5
int queue[MAX];
int rear = 0;
void enqueue(int data)
{
    if(rear == MAX - 1)
        return;
    queue[rear++] = data;
}
int main()
{
    enqueue(10);
    printf("element enqueued successfully");
    return 0;
}
```

Submitted

Test Input

stdin input (optional)

Output

```
element enqueued successfully
```

Pending manual review

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

Q7 **Queue - Deque** 5 marks

Find the error in this Deque insertFront operation.
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#define MAX 5
int deque[MAX];
int front = 0;
int rear = -1;
void insertfront(int data)
{
    if ((front == 0 && rear == MAX -1) || front == rear + 1)
    {
        printf("deque full");
        return;
    }
    if (front == 0)
        front = MAX - 1;
    else
        front = front - 1;
    deque[front] = data;
}
int main()
{
    insertfront(10);
    printf("element inserted at front");
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
deque fullelement inserted at front
```

Pending manual review

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

Q1	Stack	5 marks
Q2	Circular Queue	5 marks
Q3	Stack	5 marks
Q4	Stack	5 marks
Q5	Doubly Linked List	5 marks
Q6	Queue	5 marks
Q7	Queue - Deque	5 marks
Q8	Binary Search	5 marks
Q9	Circular Queue	5 marks
Q10	Stack - Postfix Expression	5 marks

10/10 answered

Q8 **Binary Search** 5 m

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid, r, x);
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

int binarySearch(int arr[], int l, int r, int x)
{
    if (l > r)
        return -1;

    int mid = l + (r - l) / 2;

    if (arr[mid] == x)
        return mid;

    if (arr[mid] > x)
        return binarySearch(arr, l, mid - 1, x);

    return binarySearch(arr, mid + 1, r, x);
}

int main()
{
    int arr[] = {10, 20, 30, 40, 50};
    int n = 5;
    int x = 30;

    int result = binarySearch(arr, 0, n - 1, x);
```

Test Input

stdin input (optional)

Output

```
Element found at index 2
```

Pending manual review

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q9 **Circular Queue** 5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

int dequeue()
{
    if (front == -1)
        return -1;

    int data = queue[front];

    if (front == rear)
    {
        front = -1;
        rear = -1;
    }
    else
    {
        front = (front + 1) % MAX;
    }
```

Test Input

stdin input (optional)

Output

```
Dequeued element: 10
```

Pending manual review

ENGINEER TO EXCEL

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

MADINENI DHANUSH KUMAR
192525318RA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- **Q1** Stack 5 marks
- **Q2** Circular Queue 5 marks
- **Q3** Stack 5 marks
- **Q4** Stack 5 marks
- **Q5** Doubly Linked List 5 marks
- **Q6** Queue 5 marks
- **Q7** Queue - Deque 5 marks
- **Q8** Binary Search 5 marks
- **Q9** Circular Queue 5 marks
- **Q10** Stack - Postfix Expression 5 marks

10/10 answered

Q10 **Stack - Postfix Expression** 5 marks

Find the bug in the evaluation of a Postfix expression.
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <ctype.h>

int stack[100];
int top = -1;

void push(int value)
{
    stack[++top] = value;
}

int pop()
{
    return stack[top--];
}

int main()
{
    char expression[] = "23+";
    int i = 0;

    while (expression[i] != '\0')
```

Test Input

stdin input (optional)

Output

```
Result = 5
```

Pending manual review